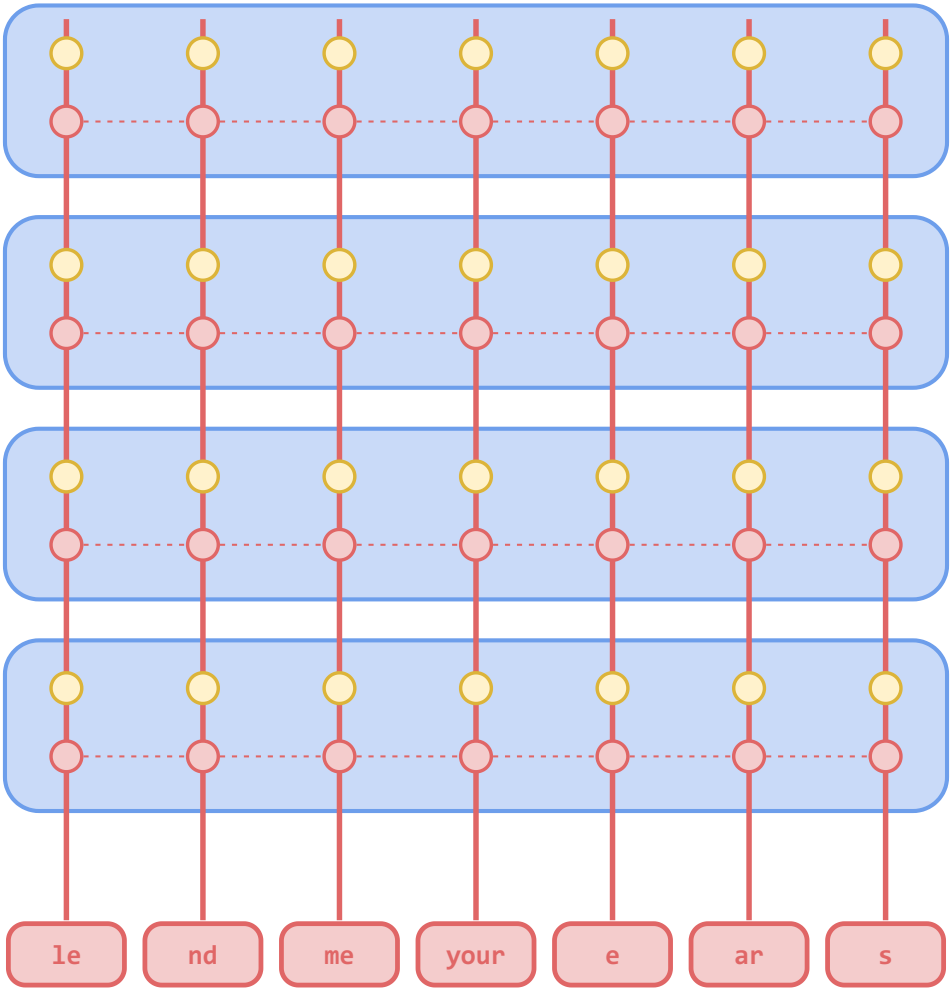
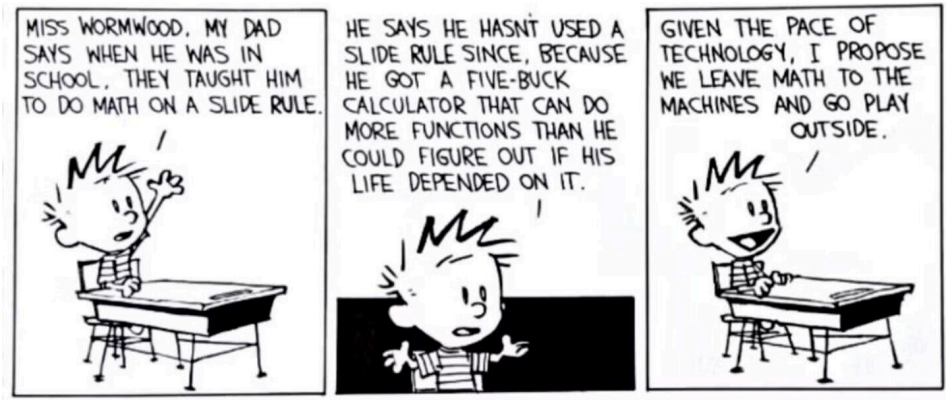


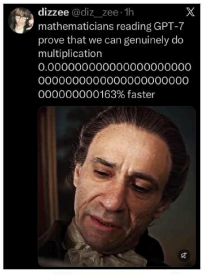

UNIT 01 · CHAPTER 12

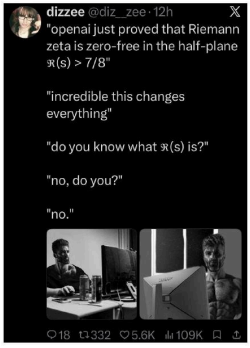
# Transformer Blocks

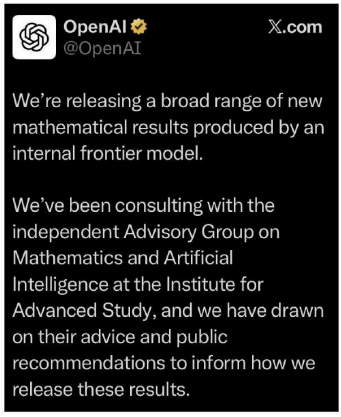
CSE 10124 · Building ChatGPT
William Theisen · University of Notre Dame

# In the News


MISS WORMWOOD, MY DAD SAYS WHEN HE WAS IN SCHOOL, THEY TAUGHT HIM TO DO MATH ON A SLIDE RULE.
HE SAYS HE HASN'T USED A SLIDE RULE SINCE, BECAUSE HE GOT A FIVE-BUCK CALCULATOR THAT CAN DO MORE FUNCTIONS THAN HE COULD FIGURE OUT IF HIS LIFE DEPENDED ON IT.
GIVEN THE PACE OF TECHNOLOGY, I PROPOSE WE LEAVE MATH TO THE MACHINES AND GO PLAY OUTSIDE.

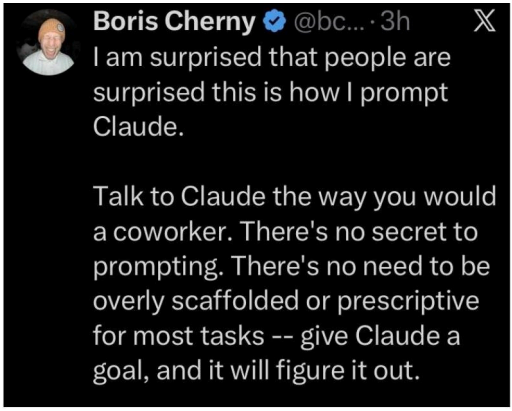

dizzee @diz_zee · 1h
mathematicians reading GPT-7 prove that we can genuinely do multiplication 0.000000000000000000000000000000000000000000000000000000000163% faster

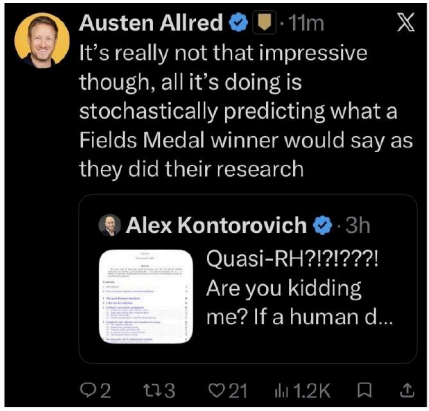

dizzee @diz_zee · 12h
"openai just proved that Riemann zeta is zero-free in the half-plane ℜ(s) > 7/8"
"incredible this changes everything"
"do you know what ℜ(s) is?"
"no, do you?"
"no."
18 332 5.6K 109K

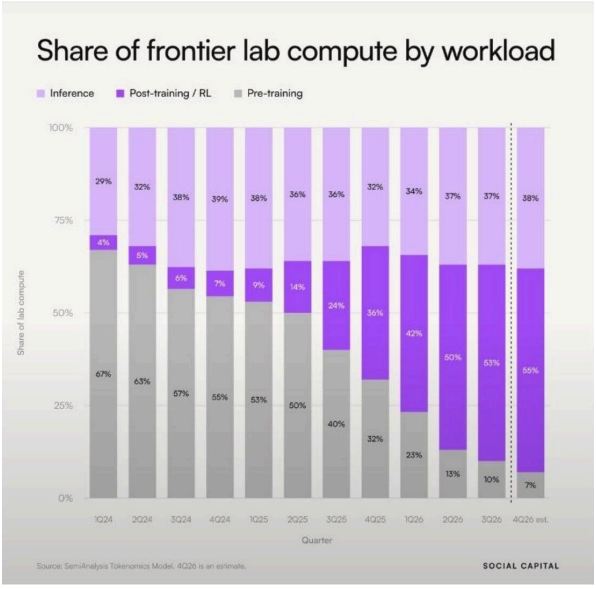

OpenAI
@OpenAI
X.com
We're releasing a broad range of new mathematical results produced by an internal frontier model.
We've been consulting with the independent Advisory Group on Mathematics and Artificial Intelligence at the Institute for Advanced Study, and we have drawn on their advice and public recommendations to inform how we release these results.


Archer @4rcherh... · 13h
Daily AI News
- US: OpenAI solves every math problem ever
- Europe: Mistral releases the worst 1T parameter model to date
6 44 1.1K 30K

# In the News


Boris Cherny @bc... · 3h
I am surprised that people are surprised this is how I prompt Claude.
Talk to Claude the way you would a coworker. There's no secret to prompting. There's no need to be overly scaffolded or prescriptive for most tasks -- give Claude a goal, and it will figure it out.


Austen Allred · 11m
It's really not that impressive though, all it's doing is stochastically predicting what a Fields Medal winner would say as they did their research
Alex Kontorovich · 3h
Quasi-RH?!?!???! Are you kidding me? If a human d...
2
3
21
1.2K

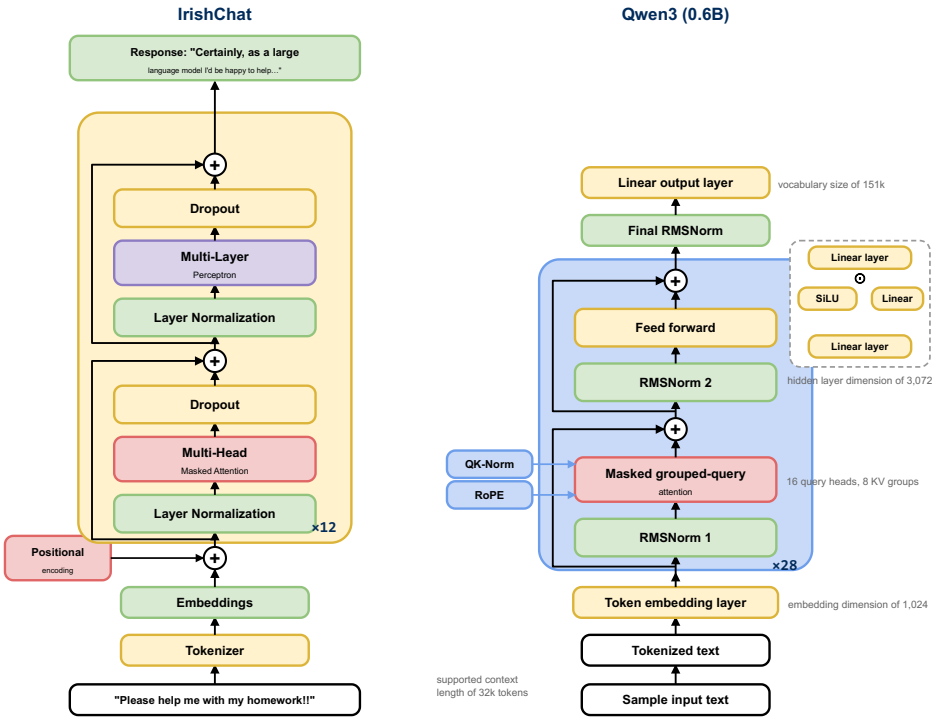

Share of frontier lab compute by workload
Inference
Post-training / RL
Pre-training
100%
75%
50%
25%
0%
Share of lab compute
29%
32%
38%
39%
38%
36%
36%
32%
34%
37%
37%
38%
4%
5%
6%
7%
9%
14%
24%
36%
42%
50%
53%
55%
67%
63%
57%
55%
53%
50%
40%
32%
23%
13%
10%
7%
1Q24
2Q24
3Q24
4Q24
1Q25
2Q25
3Q25
4Q25
1Q26
2Q26
3Q26
4Q26 est.
Quarter
Source: SemiAnalysis Tokenomics Model. 4Q26 is an estimate.
SOCIAL CAPITAL

# In the News

Investors who SpaceX has previously approached about financing its multibillion-dollar chip purchase said they only received a short two-page deal memo with pictures of outer space and an arrow pointing out that the company was going to build data centres "somewhere in the universe".

"How are we supposed to take that to the IC?" one of the people said, referring to the in-house investment committee that approves transactions.

# CSE 10124: Building ChatGPT




**New today:** everything inside the transformer block.

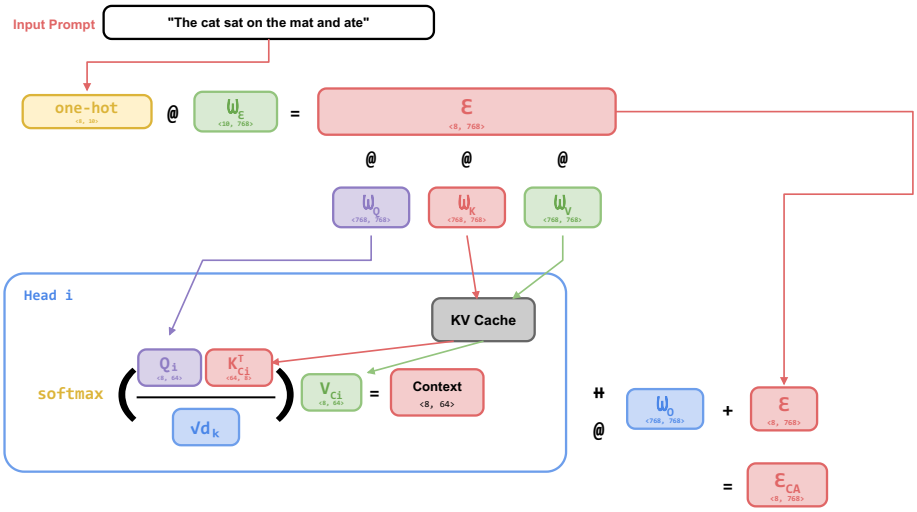
# Differences between IrishChat and Qwen3

- Multi-Head vs. GQA
- ReLU vs. SiLU
- QKNorm
- LayerNorm vs. RMSNorm
- Positional Encoding vs. RoPE
- Dropout vs. No dropout

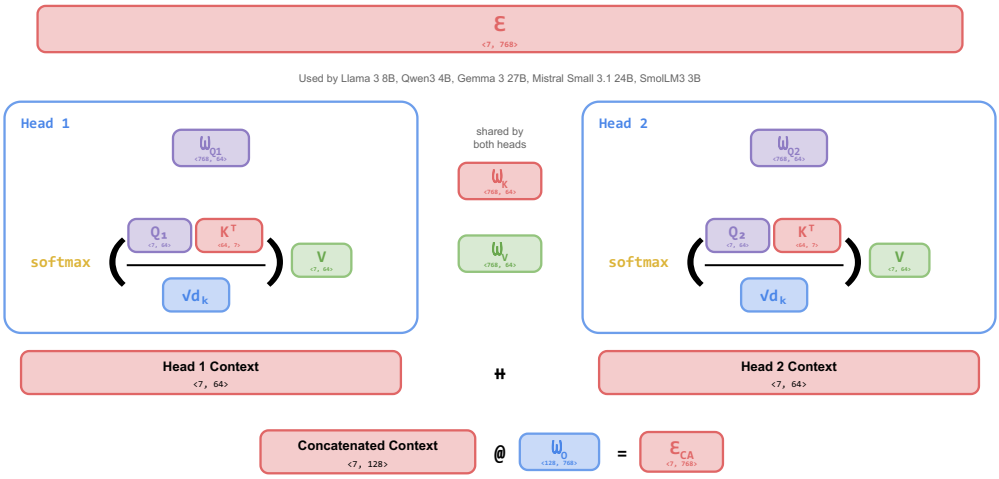
# Differences: Multi-Head vs. GQA

Difference 1 of 6

# Multi-Head Attention with KV Cache



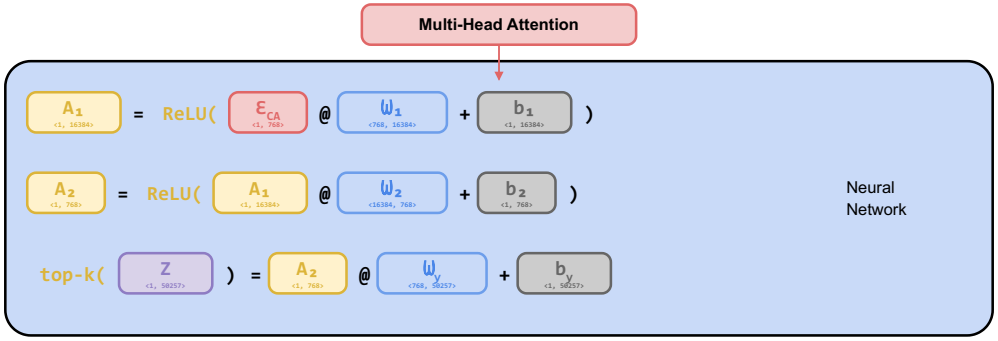
# Variant: Group-Query Attention (GQA)



# Differences: ReLU vs. SiLU

Difference 2 of 6

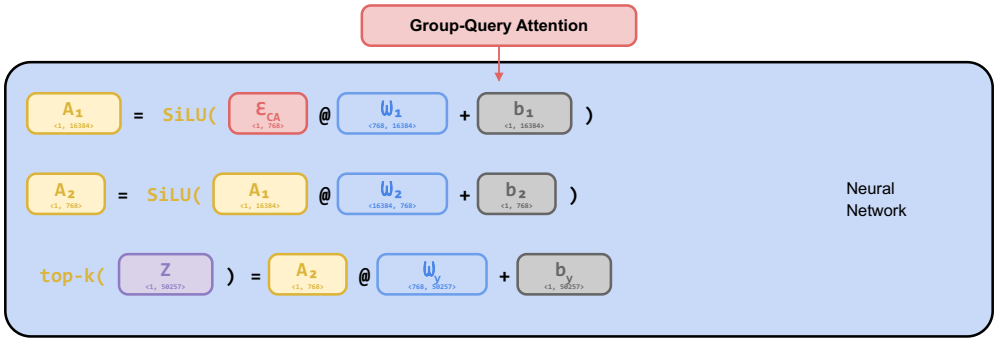
# Producing the output prediction




| softmax(Z) | 0.1739 | 0.1079 | 0.0981 | 0.0779 | 0.0629 |
|---|---|---|---|---|---|
| Token ID | **262** | 257 | 465 | 13 | 607 |
| Token | **" the"** | " a" | " his" | "." | " her" |

| softmax(Z) | 0.0614 | 0.0393 | 0.0241 | 0.0223 | 0.0116 |
|---|---|---|---|---|---|
| Token ID | 663 | 11 | 340 | 617 | 319 |
| Token | " its" | "," | " it" | " some" | " on" |

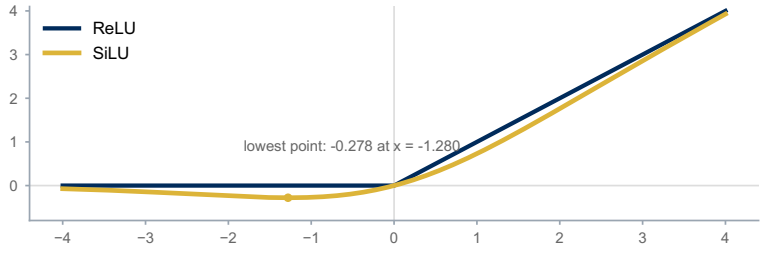
# Producing the output prediction




| softmax(Z) | 0.1739 | 0.1079 | 0.0981 | 0.0779 | 0.0629 |
|---|---|---|---|---|---|
| Token ID | **262** | 257 | 465 | 13 | 607 |
| Token | **" the"** | " a" | " his" | "." | " her" |

| softmax(Z) | 0.0614 | 0.0393 | 0.0241 | 0.0223 | 0.0116 |
|---|---|---|---|---|---|
| Token ID | 663 | 11 | 340 | 617 | 319 |
| Token | " its" | "," | " it" | " some" | " on" |

# ReLU vs. SiLU




$$[12.1]\ \mathrm{SiLU}(x) = x \cdot \sigma(x)$$

# Differences: LayerNorm vs. RMSNorm and QKNorm

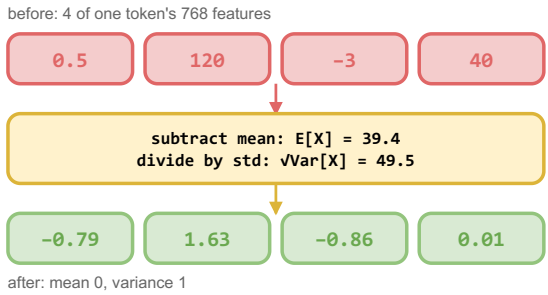
Differences 3 and 4 of 6

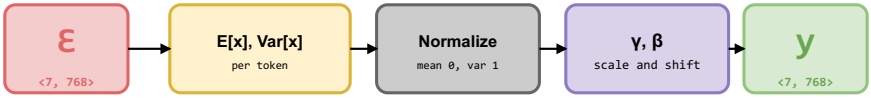
# Layer Normalization: LayerNorm

## Keeps every token's features on a stable scale

ε is a tiny constant (e.g. 1e-5) so we never divide by zero

$$[12.2] \quad y = \frac{x - E[x]}{\sqrt{Var[x] + \epsilon}} * \gamma + \beta$$



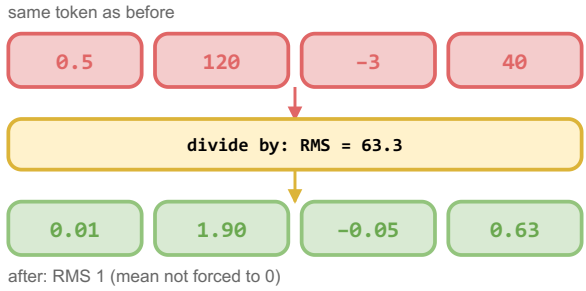




- **Scale drifts:** residual adds pile up, so activations grow with depth
- **Stable inputs:** huge values saturate softmax and kill gradients
- **Faster training:** smoother loss, higher learning rates work
- **Per token, not per batch:** same at train and inference, any batch size
- **γ, β:** learned scale and shift keep the layer expressive

# RMSNorm

A simpler **LayerNorm** that most modern **LLMs** use instead

$$[12.3]\quad y = \frac{x}{\sqrt{\text{mean}(x^2) + \epsilon}} * \gamma$$

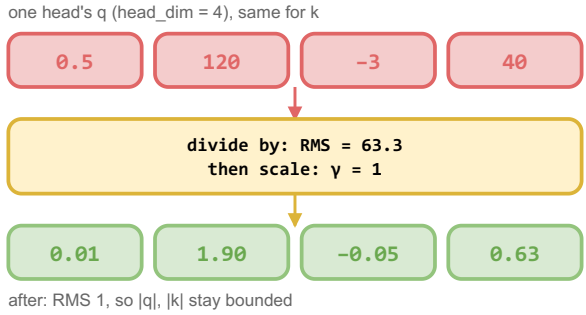



- **Drops centering:** no mean subtraction, no β
- **Cheaper:** one statistic instead of two
- **Same quality:** rescaling matters more than recentering

# QKNorm: RMSNorm on Q and K

Normalize each head's **Q** and **K** before the dot product, so **attention scores** can't blow up

$$[12.4]\quad q' = \frac{q}{\sqrt{\text{mean}(q^2) + \epsilon}} * \gamma_q$$




- **Per head:** RMSNorm over `head_dim` on **Q** and **K**
- **Why:** stops $|Q \cdot K|$ from growing, so `softmax` can't saturate
- **Where:** after the **Q**/**K** projections, before **RoPE**, **V** untouched

# Differences: Positional Encoding vs. RoPE

Difference 5 of 6

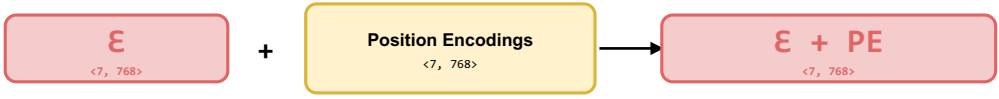
# Positional Encodings

**Attention** is order-blind, so we add a **position vector** to each token

[12.5] $PE(pos, 2i) = \sin(pos/10000^{2i/d})$

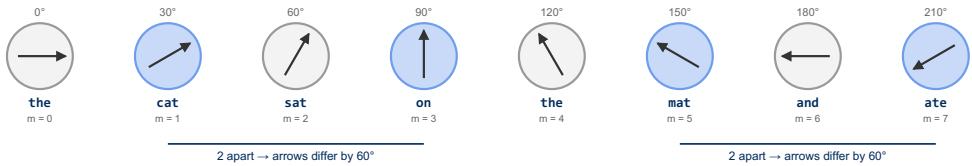
pos = token position, i = dimension pair, d = 768. Low i: fast waves. High i: slow waves.

$PE(pos, 2i + 1) = \cos(pos/10000^{2i/d})$

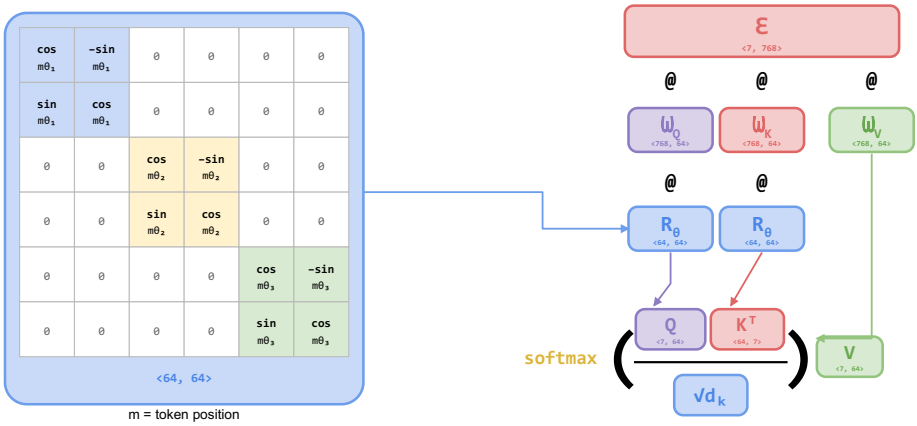


# Rotary Positional Embeddings (RoPE)

**Instead of transforming the embeddings** themselves, we transform the **Q** and **K** matrices during **attention** calculations.




This is one of the trickiest things I've seen in the past few years



# Differences: Dropout vs. No Dropout

Difference 6 of 6

# Dropout vs. No Dropout

**Dropout** **randomly zeroes activations during training** so the network can't rely on any single feature. It is only used during training!




- **Training only:** fresh random mask each step, identity at inference
- **Rescale by 1/(1 − p):** keeps the expected activation unchanged
- **Why:** helps prevent overfitting

**Dropout** is used to **fight overfitting**. However, because modern **LLMs** are trained on such an insane scale of data and the data is so widely varied, **overfitting** is much less of a concern, so dropout isn't necessary.